Market Regime Discovery in FAANG Stocks Using Feature-Engineered K-Means Clustering

Yash Jha, Param Kotadia, Shubham Raiyani, Malani Khushi
Department of Computer Science and Engineering
Unitedworld Institute of Technology, Karnavati University
Ahmedabad, India
{yashjha0117, paramkotadiya84, shubhamraiyani05, malanikhushi01}@gmail.com

***Abstract*—Stock market behavior analysis is challenging due to the highly volatile, noisy, and non-linear nature of financial data. This data is further influenced by investor sentiment, economic events, and market uncertainty. This project presents a machine learning framework that analyzes stock market behavior using the FAANG stock dataset. Historical stock attributes such as Open, High, Low, Close, and Volume are converted into useful technical features including Daily Return, Volatility, Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD). These new features better represent market conditions than raw stock prices alone. K-Means clustering is then used to group similar market states into meaningful clusters. The resulting clusters can be understood as bullish, stable, and bearish market regimes. This approach shows how unsupervised learning can uncover hidden patterns in stock market behavior for financial analysis.**

***Index Terms*—FAANG dataset, K-Means clustering, Market behavior analysis, Technical indicators, Unsupervised machine learning, Financial analytics**

I. Introduction

The stock market is regarded as one of the most complex financial systems in the modern world. Stock price movement is influenced by multiple uncertain factors such as investor psychology or sentiment, company performance, economic conditions, and global financial events. Due to the interaction of these dynamic variables, stock market data is often highly volatile, noisy, and difficult to interpret using only raw price values.

As a result, it becomes challenging to analyze market behavior effectively through direct observation of stock prices alone. Therefore, it is necessary to use alternative methods of interpretation that can provide deeper insight into how stock prices behave over time. Such methods can help in understanding hidden market patterns and support better analytical interpretation of stock market conditions [12], [14].

In this study, a feature-driven unsupervised learning framework is developed to analyze stock market behavior using FAANG stock data. Instead of directly applying clustering on raw stock prices, the proposed methodology first transforms the data into more informative technical features that better represent market conditions and short-term behavior. These engineered features allow the clustering algorithm to distinguish between different market states more effectively.

The methodology adopted in this work includes data collection, preprocessing, feature engineering, normalization, optimal cluster selection, and K-Means clustering. The resulting clusters are then interpreted to identify meaningful stock market states, such as bullish, stable, and bearish conditions. This makes the proposed framework useful not only for exploratory market analysis but also for understanding broader stock market structure.

The major contributions of this research are summarized as follows:

- Use of FAANG stock market data for behavior-based analysis
- Use of engineered financial indicators instead of raw price values
- Application of K-Means clustering for unsupervised discovery of stock market states
- Interpretation of stock market clusters as meaningful market regimes such as bullish, stable, and bearish

This research contributes by combining unsupervised clustering with feature engineering to reveal hidden behavioral patterns within FAANG stock data [13], [14]. Rather than predicting the immediate next market movement, the study focuses on understanding the underlying structural behavior of the market through clustering based on enriched technical characteristics.

II. Dataset Description

The dataset used in this project consists of historical stock market data of the **FAANG** companies, namely **Facebook (Meta), Apple, Amazon, Netflix, and Google (Alphabet)**. These companies were selected because they represent some of the most influential and actively traded technology stocks in the market, making them highly suitable for behavior-based financial analysis.

The dataset contains daily stock market records that are commonly used for market analysis and financial modeling. The original stock attributes available in the dataset are:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

Although these raw stock attributes provide important market information, they do not directly reveal hidden behavioral structure in a form suitable for clustering. Therefore,

the dataset was further transformed into a more informative feature-rich representation using selected technical and statistical indicators.

These transformed features were used as the primary input for clustering because they better capture **momentum, trend, and price variability** compared to raw stock prices alone. This makes the dataset more appropriate for unsupervised learning and stock market behavior analysis.

A. Dataset Summary

TABLE I
SUMMARY OF THE FAANG STOCK DATASET

Parameter	Description
Dataset Type	Historical Stock Market Data
Companies Covered	Facebook, Apple, Amazon, Netflix, Google
Sector	Technology
Raw Features	Open, High, Low, Close, Volume
Learning Type	Unsupervised Learning
Model Used	K-Means Clustering
Purpose	Stock Market Behavior Analysis

III. PROPOSED MODEL

The proposed model is developed to identify hidden patterns in stock market behavior using an unsupervised learning framework. In contrast to conventional stock market approaches that primarily emphasize price forecasting, this work focuses on **market regime discovery** through clustering-based behavioral analysis.

The framework converts raw historical stock market data into a more informative feature space by incorporating technical indicators and engineered financial attributes. K-Means clustering is then applied to group structurally similar market conditions based on their statistical and behavioral characteristics. This enables the identification of meaningful and interpretable market regimes, such as bullish, stable, and bearish conditions.

The overall workflow consists of the following stages:

- Feature selection and engineering
- Feature extraction from stock data
- Data standardization
- K-Means clustering
- Cluster evaluation

A. Features of the Proposed Model

The effectiveness of clustering depends heavily on the features used to represent the stock market data. Since raw stock prices alone are not sufficient to explain actual market behavior, the model uses a combination of original stock attributes and engineered technical indicators. Recent financial ML studies also note that carefully chosen technical features can improve how market structure is represented, though feature selection must be done thoughtfully.

The major features used in this project are:

Primary Stock Features

These are directly obtained from the dataset:

Open – opening price of the stock

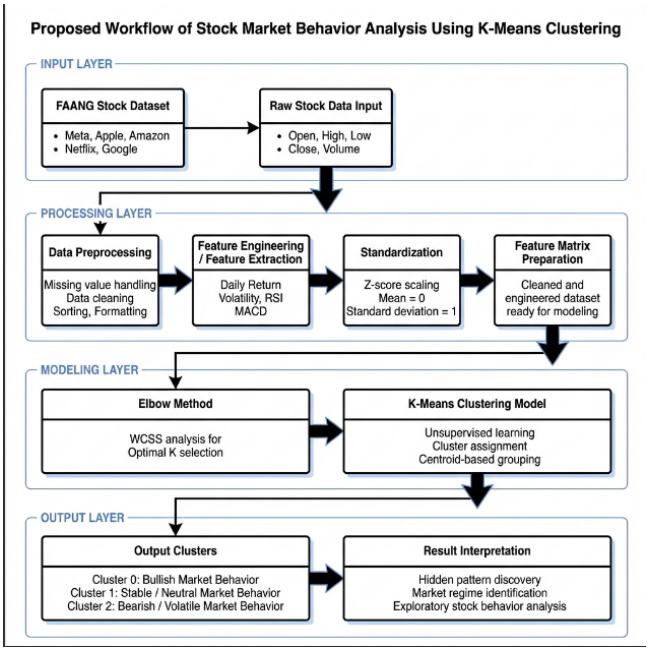

Fig. 1. Proposed workflow of stock market behavior analysis using K-Means clustering

High – highest trading price of the day

Low – lowest trading price of the day

Close – closing price of the stock

Volume – total trading volume

Engineered Technical Features

These features are derived from the raw stock data to better represent market behavior:

Daily Return

Volatility

RSI (Relative Strength Index)

MACD (Moving Average Convergence Divergence)

These features were selected because they capture important financial properties such as:

Price movement

Trend direction

Momentum

Risk and instability

Thus, the chosen features help the model represent stock market behavior more meaningfully than raw prices alone.

B. Feature Extraction

Feature extraction is an integral part of the proposed model. It is the process by which the historical data related to the stock market is converted into useful numerical data. Instead of directly using the price-related columns, the model creates other relevant features. These features describe the behavior of the stock market. Studies related to the analysis of the stock market describe the transformed data to effectively describe the behavior related to the trends, risk, and momentum [13], [15].

The features extracted by the model can be described as

follows:
(i) Daily Return
Daily Return is the percentage change in the closing price over the course of two successive days. It is the change in the closing price. It is used to determine the gain or loss in the value of the stocks.
(ii) Volatility
Volatility is the standard deviation of the returns over a certain period. It is the measure of the risk involved in the stocks. High volatility indicates high risk in the market.
(iii) RSI (Relative Strength Index)
RSI is the momentum indicator. It is used to determine the overbought and oversold conditions. It is the measure to understand the short-term market behavior.
(iv) MACD (Moving Average Convergence Divergence)
MACD is the measure to understand the trends and the momentum. It is the convergence and divergence of the moving averages.

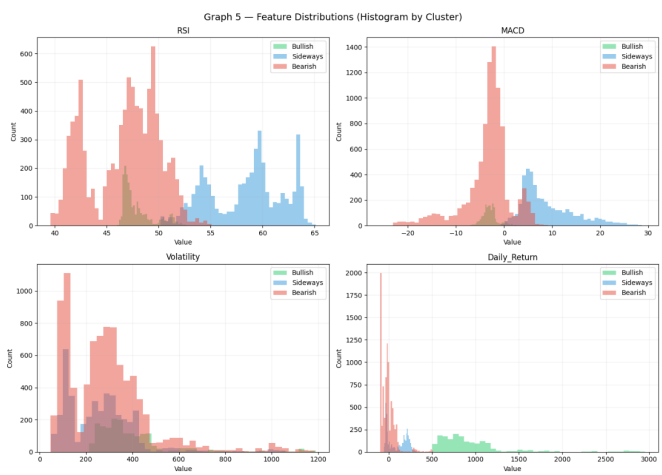

Fig. 2. Feature distribution of engineered stock market indicators across bullish, sideways, and bearish market clusters.

C. Standardization

The selected features are then standardized before applying the clustering algorithm, to ensure that all features contribute equally to the clustering process. This is done because, in the dataset, some features have different ranges of numerical values. For example, Volume might be in millions, whereas RSI will be within a range of 0 to 100. This will affect the clustering because, in K-means, distance calculations are done, and different ranges of values will affect clustering.

As mentioned in the Scikit-learn library documentation and other clustering algorithm references, distance-based clustering is scale-dependent.

To avoid such a problem, standardization is applied to the dataset, also known as z-score scaling, using the following equation:

$$Z = \frac{X - \mu}{\sigma} \tag{1}$$

Where:

- X = value of a feature
- μ = mean of a feature
- σ = standard deviation of a feature

As a result of applying standardization to the dataset, we will have:

each feature's mean = 0 each feature's standard deviation = 1

This ensures that no feature affects clustering more than another, thus making clustering more fair and quality-oriented. This approach was also adopted in a recent clustering algorithm-based finance-oriented research paper [12], [15].

D. Model Description

The principal model used in this project is K-Means Clustering Algorithm, which is a type of unsupervised machine learning method. K-Means is a suitable choice for this project since there are no output labels in the dataset, such as "bullish" or "bearish." Rather, this project aims to identify similar market behaviors using historical stock prices.
Why K-Means is Used
The K-Means clustering method is used in this project for the following reasons:
The dataset is unlabeled.
The goal is pattern discovery, not prediction.
It is appropriate for grouping similar data.
It is simple to implement.
It is simple to interpret.

Working of K-Means
The K-Means clustering method operates in the following manner:
Choose number of clusters K.
Randomly initialize centroids of clusters.
Assign data points to their respective nearest centroids.
Recalculate new centroids.
Repeat until convergence.

In this project, the Elbow method is used to determine the value of K, and finally, K-Means clustering groups the stock market data into three meaningful market behaviors:
Cluster 0: Bullish market behaviors.
Cluster 1: Stable/Neutral market behaviors.
Cluster 2: Bearish/Volatile market behaviors.

Thus, this proposed model transforms large amounts of historical stock market data into meaningful and interpretable market behaviors.

E. Performance Evaluation

Since K-Means is an unsupervised learning algorithm, supervised evaluation metrics such as accuracy, precision, recall, and F1-score are not applicable. Therefore, the proposed model is evaluated using clustering quality measures that assess how well the data points are grouped.

The major evaluation criteria are:

(i) WCSS (Within-Cluster Sum of Squares)
WCSS measures the squared distance between data points and their respective cluster centroids. Lower WCSS indicates more compact clusters.

(ii) Elbow Method
The Elbow Method is used to determine the optimal number of clusters by analyzing the rate of decrease in WCSS for different values of K.

(iii) Cluster Compactness
Cluster compactness indicates how closely related the data points are within the same cluster.

(iv) Cluster Separation
Cluster separation measures how distinctly different one cluster is from another.

(v) Interpretability
An effective clustering model should produce meaningful groups that can be interpreted as stock market regimes such as bullish, stable, and bearish behavior.

F. Figures and Tables

TABLE II
DATASET SUMMARY STATISTICS

Feature	Mean	Max
Open	144.09	782.98
High	145.74	795.06
Close	144.15	784.06
Volume	1.17e8	1.65e9
RSI	55.63	99.42
Daily Return	0.0013	0.113
Volatility	0.0185	0.118

The dataset summary gives a clear idea about how the stock features are distributed. It can be seen that the values of Open, High, and Close prices are quite similar on average, which shows that stock prices do not fluctuate drastically within a single day.

One important observation is that the Volume values are much higher compared to other features. This is expected, as trading volume is usually in large numbers and varies significantly from day to day.

The RSI values fall within the expected range of 0 to 100, which confirms that it is correctly capturing the market momentum. Daily Return values are relatively small, indicating that stock prices generally change gradually rather than suddenly.

Overall, the dataset contains features with very different ranges, which makes it necessary to apply standardization before performing clustering.

The correlation matrix helps in understanding how different features are related to each other. It is clearly visible that Open, High, Low, and Close prices are strongly correlated. This means they move together and represent similar information about stock price behavior.

TABLE III
FEATURE CORRELATION MATRIX (PARTIAL)

Feature	Open	Volume	RSI
Open	1.00	-0.41	-0.02
Volume	-0.41	1.00	-0.22
RSI	-0.02	-0.22	1.00

On the other hand, Volume shows a negative relationship with price features. This suggests that higher trading activity does not always mean higher prices, which is quite common in real stock markets.

The RSI feature shows weak correlation with price-related features, which indicates that it captures market momentum rather than actual price values.

Overall, the dataset contains a mix of strongly related and independent features, which is useful for clustering as it helps in identifying different patterns in market behavior.

RESULTS

The proposed K-Means clustering framework was applied to the preprocessed and standardized FAANG stock dataset to identify underlying market behavior patterns. The clustering results demonstrate that the selected technical indicators are effective in capturing structural differences in stock market conditions, leading to meaningful separation of data into distinct behavioral groups.

Since the model is based on unsupervised learning, the results are evaluated using clustering quality measures such as within-cluster compactness (WCSS) and visual separability of clusters, rather than traditional classification metrics.

Elbow Method Analysis

The Elbow Method was used to determine the optimal number of clusters K. This method evaluates clustering performance by analyzing the variation of inertia (Within-Cluster Sum of Squares) across different values of K.

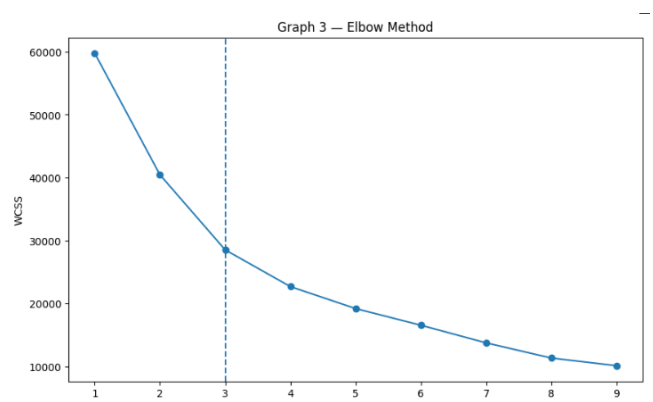

Fig. 3. Elbow method used to determine the optimal number of clusters for K-Means clustering.

Figure 3 shows that the WCSS decreases rapidly as the number of clusters increases from 1 to 3, indicating significant improvement in cluster compactness. Beyond $K = 3$, the rate of decrease in WCSS becomes considerably smaller, suggesting diminishing returns in clustering performance.

Observations

- A sharp reduction in WCSS is observed for $K = 1$ to $K = 3$, indicating that additional clusters significantly improve data representation in this range.
- For $K > 3$, the reduction in WCSS becomes gradual, implying that additional clusters do not provide substantial improvement.
- The curve exhibits a clear "elbow" at $K = 3$, which indicates the most suitable number of clusters for the dataset.

Inference

Based on the Elbow Method analysis, the optimal number of clusters is determined as $K = 3$. This result aligns with the presence of three distinct market regimes within the dataset.

The clustering output confirms that the stock market behavior can be effectively categorized into three meaningful states:

- Bullish market behavior
- Stable / neutral market behavior
- Bearish / volatile market behavior

This validates that the proposed feature-engineered representation successfully captures key characteristics of market dynamics and enables interpretable segmentation of stock market conditions [12], [13].

IV. K-Means Clustering and Cluster Interpretation

Once the optimal value of K was determined using the Elbow Method, the K-Means clustering algorithm was applied to the engineered stock market feature set. The objective of clustering was to group stock market data points into distinct categories based on similarity in market behavior rather than predefined labels.

The clustering process was performed using the selected normalized features:

- Daily Return
- Volatility
- Relative Strength Index (RSI)
- Moving Average Convergence Divergence (MACD)

These features were chosen because they capture important market characteristics such as trend, momentum, price fluctuation, and return behavior. By clustering the data in this feature space, the model was able to identify hidden structural patterns in historical stock market data.

A. Cluster Visualization Analysis

The cluster visualization provides a graphical representation of how the K-Means algorithm has separated the stock market data into different groups based on feature similarity.

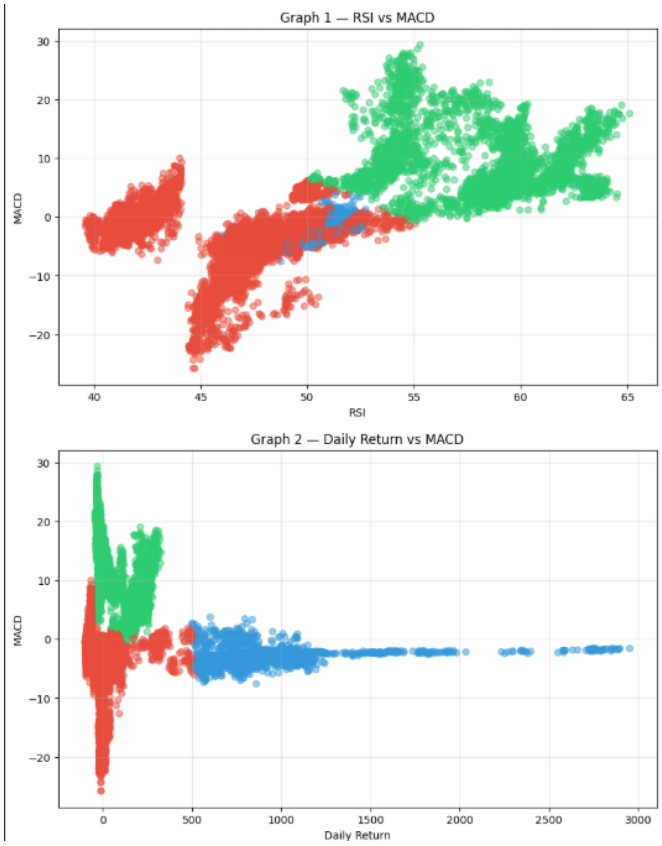

Fig. 4. Cluster visualization using RSI vs MACD and Daily Return vs MACD showing separation of bullish, sideways, and bearish market conditions.

Figure 4 presents the clustering results using two feature combinations: RSI vs MACD and Daily Return vs MACD. It can be observed that the data points are grouped into three visually distinct clusters, each representing a different stock market condition.

The visualization confirms that the selected features are effective in distinguishing different market behaviors. Although some overlap exists between clusters, which is expected in complex financial datasets, the overall separation is sufficiently clear to support meaningful cluster interpretation.

B. Observations

The following key observations can be made from the clustering results:

- The stock market data points are grouped into three major clusters.
- Each cluster contains data points with similar values of Daily Return, Volatility, RSI, and MACD.
- Some overlap exists between the clusters due to the naturally noisy and dynamic nature of financial market data.
- The clusters still show meaningful separation, indicating that the selected features are useful for stock market behavior analysis.

C. Interpretation of Clusters

The final clusters were interpreted based on the financial behavior of the selected features. Each cluster represents a meaningful stock market regime.

1) Cluster 0 – Bullish Market Behavior: This cluster represents a bullish market state, where the market generally shows positive movement and stronger upward momentum.

Characteristics:

- Higher Daily Return
- Stronger positive trend movement
- Relatively stable market behavior

Interpretation: This cluster reflects a growth-oriented market condition in which stock prices tend to move upward with comparatively favorable momentum.

2) Cluster 1 – Stable / Neutral Market Behavior: This cluster represents a stable or sideways market condition, where stock movement is moderate and less directional.

Characteristics:

- Moderate Daily Return
- Controlled Volatility
- Neutral or balanced trend movement

Interpretation: This cluster reflects a relatively balanced market condition in which stock prices move within a moderate range without strong bullish or bearish momentum.

3) Cluster 2 – Bearish / Volatile Market Behavior: This cluster represents a bearish or high-risk market condition, where the market shows instability and weaker performance.

Characteristics:

- Lower Daily Return
- Higher Volatility
- Unstable or negative trend movement

Interpretation: This cluster reflects a risky market state associated with downward pressure, higher uncertainty, and unstable stock behavior.

D. Separation of Market States

The clustering results show that stock market behavior can be separated into different market states using engineered technical features and unsupervised learning. This supports the idea that historical stock data contains hidden structural patterns that can be uncovered through clustering.

The separation of market states in this work is primarily achieved through:

- Feature Engineering
- Data Standardization
- K-Means Clustering

The identified market states are mainly distinguished based on:

- Return Behavior
- Volatility Level
- Market Momentum
- Trend Strength

As a result, the proposed framework successfully groups the stock market data into three practical and meaningful states:

- Bullish
- Stable / Neutral
- Bearish / Volatile

V. Performance Evaluation

A. Feature Profile Analysis

To further evaluate the effectiveness of the clustering results, a feature profile analysis was performed by calculating the average values of key features within each cluster.

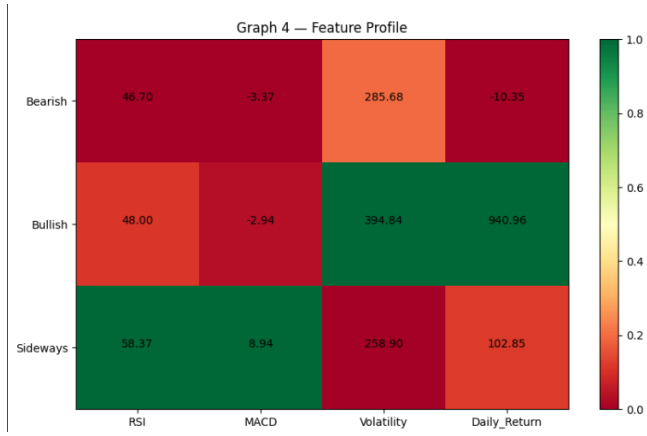

Fig. 5. Feature profile heatmap showing average feature values across different market clusters.

Figure 5 presents a heatmap of the average feature values for each cluster. It can be observed that each cluster exhibits a distinct pattern across the selected features.

The bullish cluster shows higher daily returns and volatility, indicating strong upward movement in the market. The sideways cluster is characterized by higher RSI and MACD values, reflecting moderate and stable market conditions. In contrast, the bearish cluster shows lower daily returns and negative MACD values, indicating downward trends and higher risk.

This analysis confirms that the clusters formed by the K-Means algorithm are not only visually separable but also statistically distinct in terms of feature behavior. Therefore, the proposed framework successfully identifies meaningful stock market regimes using unsupervised learning.

B. Discussion

The clustering results indicate that the proposed framework is effective for identifying underlying market regimes in stock data. By focusing on pattern discovery rather than prediction, the model captures structural variations in market behavior using unsupervised learning.

The use of engineered technical features enhances the representation of market dynamics, enabling K-Means to form distinct and interpretable clusters. The resulting clusters correspond to meaningful market conditions, demonstrating that the selected features effectively capture variations in trend, volatility, and momentum.

Key Findings

The Elbow Method identifies three as the optimal number of clusters, suggesting the presence of three dominant market regimes in the dataset.
Cluster visualization reveals clear separation between market states, supporting the effectiveness of the feature space in distinguishing different behaviors.
Technical indicators contribute significantly to clustering quality by capturing key financial properties such as trend direction, risk, and momentum.

Conclusion from Results

The clustering results confirm that K-Means can effectively identify latent structures in stock market data. The model provides a structured representation of market behavior, enabling classification of market conditions into bullish, stable, and bearish regimes without relying on labeled data.

C. Limitations

Despite its effectiveness, the proposed framework has certain limitations. The K-Means algorithm assumes that clusters are formed around centroids and are relatively compact in shape, which may not always reflect the complex and non-linear structure of financial market data. In addition, stock market behavior is naturally noisy and dynamic, which can reduce the clarity of cluster boundaries.

Another limitation is that the proposed model relies only on historical stock prices and technical indicators. It does not incorporate external variables such as macroeconomic conditions, market news, investor sentiment, or global events, all of which can significantly influence stock market behavior.

Therefore, the results of this work should be interpreted as an exploratory analysis of market regimes rather than a complete representation of all real-world market influences [14], [15].

VI. Conclusion

This study presented a feature-driven unsupervised learning framework for stock market behavior analysis using FAANG stock data. Instead of relying only on raw stock prices, the proposed approach transformed the historical stock records into more informative technical features such as Daily Return, Volatility, RSI, and MACD. These engineered features provided a more meaningful representation of market behavior for clustering analysis.

The K-Means clustering algorithm was applied after preprocessing, feature engineering, normalization, and optimal cluster selection using the Elbow Method. The clustering results showed that the stock market data could be effectively grouped into three meaningful market states: **bullish, stable/neutral, and bearish/volatile**. The obtained clusters were further supported through visual analysis, feature distributions, boxplots, scatter plots, and feature profile heatmaps.

The results demonstrate that unsupervised learning can be successfully used to discover hidden structural patterns in stock market data without relying on predefined labels. The proposed framework improves the interpretability of financial market behavior and provides a structured approach for exploratory market regime analysis.

Although the present work focuses on clustering-based market understanding rather than direct price prediction, it establishes a strong foundation for future research. In future work, this framework can be extended by integrating additional technical indicators, deep learning models, hybrid clustering approaches, or real-time stock market analysis systems.

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